GDG WINDSOR · BUILD WITH AI · EDGE / ON-DEVICE TRACK

Gemma Without Borders

An autonomous study agent for the Grade 9 EQAO math assessment.

Runs entirely on the student's device. A child's mistakes never leave their machine.

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The problem: wrong answers are symptoms, not causes

- Every Ontario Grade 9 student writes the EQAO math assessment. Many arrive with **specific, well-documented misconceptions** — not general weakness.
- A student who answers $\frac{2}{3} + \frac{1}{4} = \frac{3}{7}$ doesn't need more practice. They need someone to notice they **add tops and bottoms straight across** — and fix exactly that.
- Teachers can't do this per-student, per-misconception. Generic review apps don't even try.

Calculate $\frac{2}{3} + \frac{1}{4}$

A) $\frac{11}{12}$ correct

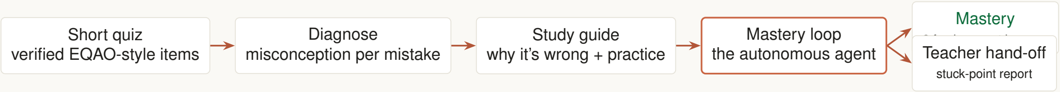
B) $\frac{3}{7}$ adds straight across

C) $\frac{3}{12}$ didn't rescale tops

D) $\frac{11}{7}$ mixed method

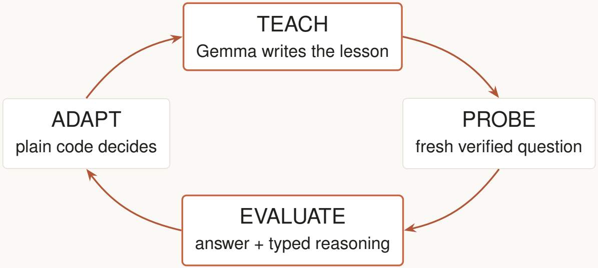
Every wrong option is the exact result of a *known* misconception — so the wrong answer itself is the diagnosis.

What we built



- Student takes a 5-question quiz. On submit, the agent reads **all** the mistakes, finds the pattern, and picks the **priority misconception.**
- Then it **pursues a goal**: keep teaching — with different strategies — until the student *demonstrates* understanding, or a human should take over.

An agent, not a chatbot



A chatbot responds once.
An agent pursues a goal:

"Two consecutive fresh questions answered correctly, with reasoning that shows the misconception is gone."

It cannot run away:

- 4-attempt cap
- 12-model-call budget
- strategy exhaustion → clean teacher hand-off

Orange boxes = Gemma. Every session provably terminates.

When a lesson fails, the agent changes how it teaches

Direct correction
state the error, show the rule

Visual walk-through
number lines, area models

Side-by-side contrast
wrong vs right, line by line

Real-world analogy
money, pizza, game scores

- Adaptation is a **fixed ladder** — every retry is a provably different pedagogy, never an open-ended improvisation.
- **Gemma chooses which rung comes next**, by reading the student's own typed reasoning — and tells the student why: "Trying a visual walkthrough — you said you can't see why the denominators matter."
- Right answer but shaky reasoning? **Doesn't count toward mastery.** Gemma

Where Gemma does the thinking

rubric: Gemma Integration, 30%

Gemma's job	Capability shown	Guardrail
Write the lesson & explanation	generation, pedagogy	grounded in the verified solution — may not invent numbers
Grade typed reasoning	reasoning, structured output	closed labels; fail-open; never punishes the student
Choose the next strategy	decision-making	restricted to the remaining ladder; deterministic fallback
Author new questions	structured output (JSON)	must pass its own blind re-solve, else discarded

- Remove Gemma and there is no tutor. **But the math truth never depends on it.**
- One auditable door (`gemma_client.py`); model size is a dial: `GEMMA_MODEL=gemma3:12b`.

The data Gemma stands on

rubric: Functionality, 20%

A verified diagnostic bank:

- 36 original EQAO-style questions across all 5 strands of the Ontario MTH1W curriculum
- 48-misconception taxonomy, each with the flawed thinking spelled out
- every wrong option tagged with the misconception that produces it
- independently re-solved and adversarially checked; 10 tags corrected in review

So the agent can be trusted:

- **Diagnosis** = table lookup on the tags. No model call, no hallucination, 100% reproducible.
- **Probes** are served bank-first — a generated question is a last resort, and must audit itself.
- **Every number a student sees** traces to a verified solution.

We caught our model failing — and engineered around it

Every guardrail exists because live testing broke something. The commit history is the proof of work.

Caught in testing (Gemma 1B)	Architectural fix
Generated a question with a **wrong answer key**	blind self-solve audit; bank-first probes
Wrote a probe **in Spanish**	constrained generation prompts
Invented wrong arithmetic in an explanation ($\frac{2}{3} + \frac{1}{4} = \frac{17}{12}$)	grounding rule: explain *why*, never recompute — all numbers must come from the verified solution
Rambled when reconstructing multi-step errors	diagnosis by ground-truth lookup, not model reasoning

- The result: **the model is never the source of truth for math** anywhere a student can see it.

Fully on-device

the Edge track promise, kept literally

- Gemma runs locally via **Ollama** — the demo works in airplane mode.
- Tutoring data is a protected student record. Here, **what a student struggles with stays on their machine** — no vendor agreement, no cloud log.
- Runs on a normal laptop: 1B for instant responses, 12B for richer teaching. Same code, one environment variable.
- No accounts, no API keys, no network calls in the loop — by construction.

The student's laptop

Streamlit app

quiz · agent loop · reports

↕

Ollama + Gemma

1B installed · 12B one flag away

nothing crosses this line

(the internet)

The 90-second live demo

1. **Quiz** — five questions, we miss the fraction item with the classic straight-across error.
2. **Results** — the agent finds the pattern, names the misconception, builds the study guide.
3. **"Practice until I've mastered it"** — lesson appears; we answer wrong *and type our flawed reasoning*.
4. **The moment**: the agent reads our words, **switches teaching strategy, and explains why** — attempt counter, approach, and streak all visible.
5. We answer two fresh questions correctly → **Mastery demonstrated**, with a recap of which approaches it took.
6. (And if we kept failing: a teacher hand-off report — the agent knows its limits.)

Everything shown runs live on this laptop. Wi-Fi can be off.

What's next

- **Photograph your written work** — Gemma 12B reads handwriting on-device; it transcribes, the verified bank judges. (vision is in the model we already ship — still zero cloud)
- **Local tool use** — Gemma emits structured tool calls; a local SymPy verifier proves every check deterministically. (tool use for a model that doesn't natively support it — via constrained JSON)
- **Game mode** — a three.js world where students move station to station; same agent brain underneath.

HOW WE SCORE OURSELVES ON THE RUBRIC

Gemma Integration 30%
4 roles, all guardrailed

Innovation 30%
goal-seeking tutor agent

Functionality 20%
live, tested, terminates

Writeup 20%
honest engineering story

Build something people will remember

not because it's AI, but because it's genuinely useful.

A student walks in confused about fractions.
Twenty minutes later they've *proven* they understand — and nothing about them ever left their laptop.

github.com/EdTechDL/gemma-without-borders · live demo + clonable notebook

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